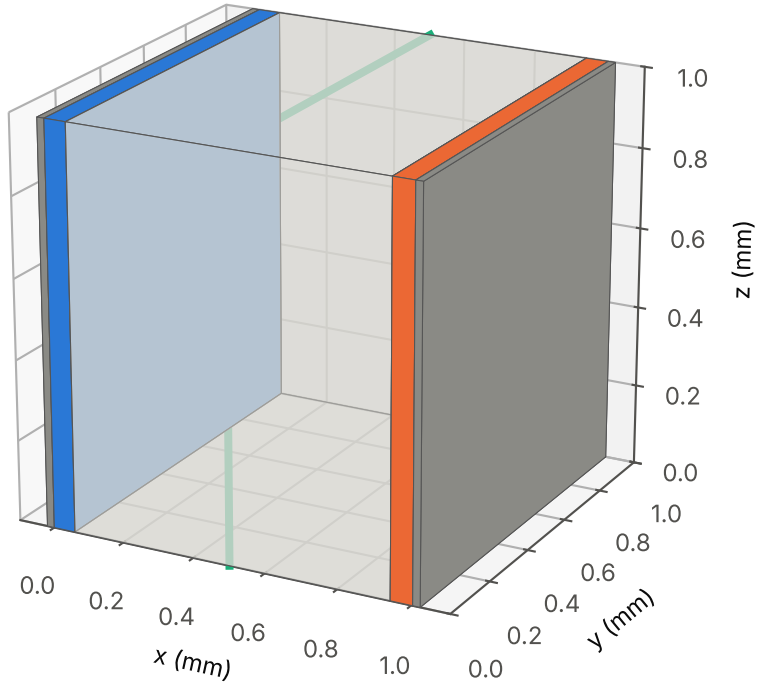
G1 PIN bar: n⁺ (2 µm) | i (1 mm) | p⁺ (2 µm), cross-section 1 mm × 1 mm

green band: heat sink at the centre (x = Xc)

n⁺ layer (blue)
+ ohmic contact (grey)
at 0 V

this end faces
the room



p⁺ layer (orange)
+ ohmic contact (grey)
at V_F

this end faces
the room

intrinsic region (light): W_i = 1 mm, N_bg = 10^{12} cm^{-3}

Doped layers drawn ~30x thicker than scale (true 2 µm). COMSOL solves this bar as a 1D model with cross-sectional area Ac = 1 mm^2 (G2).